INDUSTRY PROJECT REPORT

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Technology Databricks | SQL | Power BI

INDUSTRY PROJECT - 2026

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Introduction

The projects focus on understanding retail business performance, comparing year over-year results using Like-for-Like analysis, and evaluating department and category performance.

Each project demonstrates the application of data analysis concepts, business intelligence techniques, data visualization, and dashboard development to present business information in a structured and understandable format.

PROJECT 1

1. Executive Summary Dashboard

Brief Project Summary

The Retail Executive Summary Dashboard provides an overview of retail business performance through key performance indicators (KPIs) and interactive visualizations.

The dashboard was designed to present important business metrics in a consolidated view, helping users understand overall performance without having to examine multiple reports separately.

Business Problem and Explanation

Retail businesses generate large amounts of sales and transaction data. Reviewing this information across multiple tables and reports can make it difficult to understand overall business performance quickly.

The objective of this project was to organize key retail performance metrics into an executive-level dashboard. This allows users to review important indicators and explore business performance through visual summaries.

Key Work Performed

• Developed an executive summary dashboard using Microsoft Power BI.

• Organized key performance indicators into a consolidated view.

• Created visualizations to present retail business metrics.

• Worked with sales and transaction-related measures.

• Used DAX to calculate and display relevant metrics.

• Designed charts and other visuals to make the dashboard easier to understand.

• Arranged dashboard elements to improve readability and navigation.

Key Insights and Findings

The dashboard provides a consolidated view of important retail performance indicators.

The analysis focuses on:

• Understanding overall sales performance.

• Monitoring transaction-related metrics.

• Reviewing key business indicators.

• Identifying trends through visual representations.

• Making important business metrics easier to interpret.

Business Impact

An executive summary dashboard helps stakeholders review important business indicators in one place.

By presenting key metrics through interactive visualizations, the dashboard reduces the need to examine multiple reports separately and makes business performance easier to communicate.

It also provides a foundation for further analysis of sales trends, customer behaviour, and other retail performance measures.

PROJECT 2

Like-for-Like (LFL) Performance Analysis

Brief Project Summary

The Retail Like-for-Like (LFL) Analysis Dashboard focuses on comparing current-year and previous-year retail performance using comparable business metrics.

The dashboard presents growth indicators that help users understand changes in sales, transactions, and Average Transaction Value (ATV), along with regional and store-format performance.

Business Problem and Explanation

Comparing business performance across different years can be challenging when the periods being compared are not directly equivalent.

Like-for-Like analysis helps evaluate performance across comparable periods, providing a more meaningful view of growth than simply looking at total sales figures in isolation.

The objective of this project was to present LFL performance indicators in an interactive dashboard and make year-over-year comparisons easier to interpret.

Key Work Performed

• Developed a Like-for-Like analysis dashboard in Power BI.

• Reviewed current-year and previous-year performance measures.

• Analysed sales growth and transaction growth.

• Included Average Transaction Value (ATV) growth analysis.

• Compared performance across regions.

• Explored differences in performance across store formats.

• Used DAX measures to present growth metrics.

• Created charts, KPI cards, and other visualizations.

• Organized the dashboard to make comparisons easier to understand.

Key Insights and Findings

The dashboard supports the analysis of retail growth across multiple performance indicators.

The main areas examined include:

• Changes in sales compared with the previous year.

• Growth or decline in transaction volume.

• Changes in Average Transaction Value.

• Differences in performance across regions.

• Performance comparisons across store formats.

• Trends in business performance over time.

These comparisons help distinguish overall growth from changes in individual performance indicators.

Business Impact

LFL analysis provides a structured way to compare business performance across equivalent periods.

Examining sales, transactions, and ATV together can help identify whether changes in sales are associated with transaction volume, average spending, or a combination of factors.

Regional and store-format comparisons can also help identify areas that may require further investigation.

PROJECT 3

Department Analytics

Brief Project Summary

The Department Analytics focuses on analysing retail performance at department and product-category levels.

The dashboard presents sales performance, category contribution, customer-related metrics, regional distribution, monthly trends, and comparisons across store formats.

It provides a more detailed view of business performance than an executive summary by examining how different categories contribute to overall results.

Business Problem and Explanation

Understanding total sales alone does not explain how individual product categories contribute to business performance.

Retail businesses need to examine category-level results, regional differences, customer metrics, and store-format patterns to understand where performance is concentrated.

The objective of this project was to organize these metrics into an interactive dashboard that supports detailed department and category analysis.

Key Work Performed

• Developed department and category performance pages in Power BI.

• Analysed sales performance across available product categories.

• Created visualizations to present category-wise sales contribution.

• Reviewed regional sales distribution.

• Examined monthly sales trends.

• Compared category performance across store formats.

• Used DAX measures to display relevant business metrics.

• Created charts, matrices, and KPI cards.

• Designed dashboard pages to support both high-level and detailed analysis.

Key Insights and Findings

The dashboard supports analysis of the following areas:

• Sales distribution across product categories.

• Contribution of individual categories to total sales.

• Regional differences in sales performance.

• Changes in sales across months.

• Category mix across different store formats.

• Customer-related performance indicators.

The category-level analysis helps identify how sales are distributed across the available categories and provides a basis for investigating differences in performance.

Business Impact

Department and category analysis helps businesses understand the composition of their sales and identify categories that contribute more or less to overall performance.

Comparing categories across regions and store formats can provide useful context for further investigation into product demand and sales patterns.

The dashboard also demonstrates how Power BI can be used to present detailed business metrics in an organized and interactive format.

PROJECT 4

Customer Analysis

Brief Project Summary

The Customer Analysis Dashboard provides an overview of key customer metrics, including total customers, new customers, repeat rate, transactions per customer, loyalty sales, and spend per customer.

The dashboard also enables analysis by region, loyalty tier, age group, gender, and monthly trends.

Business Problem and Explanation

Understanding customer behaviour requires more than simply tracking the total number of customers.

Businesses need to evaluate customer engagement, repeat purchasing patterns, customer value, and differences across customer segments.

The objective of this project was to bring these metrics together in an interactive dashboard, enabling users to monitor customer trends, compare regional performance, understand loyalty-tier contributions, and analyse customer demographics.

Key Work Performed

Analysed key customer metrics, including customer count, new customers, repeat rate, and transactions per customer.

Compared customer counts and spend per customer across different regions.

Examined customer share and sales contribution across Gold, Platinum, and Silver loyalty tiers.

Analysed customer distribution and spending-related metrics across different age groups and gender.

Reviewed monthly customer growth and repeat-rate trends.

Used KPI cards, charts, and tables to present customer information in a structured and understandable format.

Key Insights and Findings

The dashboard displays approximately 2K current-year customers and 2K current-year new customers.

The headline current-year repeat rate is 97.4%, with 27.31 transactions per customer displayed in the KPI cards.

Monthly customer numbers increased from 1,128 in June 2025 to 1,482 in December 2025.

The monthly repeat rate fluctuated between 83.8% and 86.5% from June 2025 to May 2026.

Among the clearly labelled age groups, customers aged 26–35 accounted for 2,122 customers (14.8%), while the 60+ age group accounted for 706 customers (4.9%).

The dashboard enables comparisons of customer distribution across regions and loyalty tiers to support further investigation of customer engagement and sales contribution.

Business Impact

The Customer Analysis Dashboard helps stakeholders monitor customer growth, engagement, and spending patterns in one place.

Regional and loyalty-tier comparisons can help identify customer segments that may require further analysis, while demographic insights provide a better understanding of customer distribution.

OVERALL PROJECT CONCLUSION

Project Summary

Project 1: Executive Summary Dashboard focused on presenting key retail performance indicators in a consolidated executive level view.

Project 2: Like-for-Like Performance Analysis Dashboard focused on comparing current-year and previous-year performance through sales growth, transaction growth, ATV growth, and other comparisons.

Project 3: Department Analytics Dashboard focused on analysing category-level sales, sales contribution, regional distribution, monthly trends, and store-format performance.

Project 4: Customer Analysis Dashboard focused on analysing customer behaviour, engagement, and value such as customer growth, repeat rate, transactions per customer and loyalty sales.

Overall Learning Outcomes

Through these projects, I gained practical experience in: • Developing interactive dashboards using Power BI.

• Working with DAX measures.

• Presenting business metrics through KPI cards and visualizations.

• Comparing business performance across periods.

• Analysing retail data at different levels of detail.

• Organizing dashboard layouts for better readability.

• Translating business questions into analytical views.

• Communicating information through data visualisation.

Overall Interpretation

Together, these projects demonstrate how business intelligence dashboards can support different levels of retail analysis.

The executive summary provides a high-level view of business performance, the LFL dashboard focuses on comparable growth, the department analytics provides a more detailed view of sales composition and the customer analysis provides a more detailed view on customer repeat rate and customer growth.

The projects helped me understand the importance of selecting relevant metrics, presenting comparisons clearly, and using visualizations to make business information easier to interpret.

Conclusion

Working on these projects strengthened my understanding of Power BI, DAX, data visualization, and business-oriented analysis. These experiences have contributed to my development as an aspiring Data Analyst and motivated me to continue improving my technical skills and analytical thinking.

Skills and Technologies Applied

Skill / Technologies	Application
POWER BI	Dashboard development
DAX Measures	Creating measure and performing calculations
Data Visualization	Understanding the business metrics through graphs and charts
KPI Analysis	Monitoring key performance indicators
Retail Analytics	Examining sales, transactions, categories and performance trends